

# 09

## Performance and Process Enhancement

A bottleneck is a mechanism, not merely a red number.

# Decompose elapsed time

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## Throughput time

Case start to completion.

## Service time

Time actively worked.

## Waiting time

Time between activities.

# Four bottleneck signatures

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- ▶ Long waiting before one activity
- ▶ Growing queue over time
- ▶ Rework returning to the same activity
- ▶ Excess handovers before completion

## Do not stop at

“Activity X has the longest average duration.”

Ask whether the delay comes from capacity, batching, policy, missing information, prioritization, or measurement.

# Compare cohorts, not only averages

| Cohort                   | Median cycle time | Rework rate | Cases |
|--------------------------|-------------------|-------------|-------|
| Standard orders          | 2.2 d             | 6%          | 4,180 |
| Custom orders            | 6.8 d             | 29%         | 640   |
| Custom + partner channel | 9.4 d             | 41%         | 185   |

## Improvement hypothesis

Partner-channel custom orders arrive with incomplete specifications, triggering clarification loops. A structured intake form should reduce rework and waiting time.

A hypothesis names a mechanism and predicts a measurable change.

# From insight to intervention

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**Key idea** Process mining supports improvement by narrowing and testing explanations. It does not establish causality by itself.

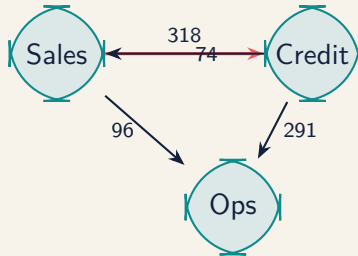
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## Organizational and Multi-dimensional Analysis

Processes are networks of coordination, not only sequences of activities.

# A handover-of-work network

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Network analysis can reveal:

- ▶ Frequent handovers and coordination loops
- ▶ Central or overloaded roles
- ▶ Segregation-of-duties concerns
- ▶ Differences between formal and actual collaboration

# Combine perspectives

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Control flow

Which path?

Time

How long?

Organization

Who coordinated?

Data

Under which conditions?

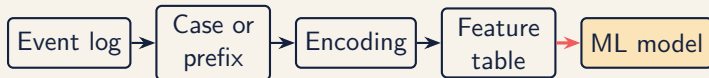
## Example synthesis

Clarification loops are slowest when cases cross from partner sales to central operations, especially for custom products with missing dimensions.



# From an event log to machine-learning rows

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## Case-level row

One completed case, usually for retrospective outcome analysis.

## Prefix-level row

One case observed up to a prediction point, used for online monitoring.

**Key idea** Define the observation cutoff before computing any feature or label.

# Encoding strategies

| Encoding       | Representation                  | Strength                    | Main risk                  |
|----------------|---------------------------------|-----------------------------|----------------------------|
| Aggregation    | Counts, durations, last value   | Simple and strong baseline  | Loses order                |
| Index-based    | Activity/attribute by position  | Preserves early order       | Wide, sparse tables        |
| Sequence       | Tokens or learned embeddings    | Captures order and context  | Data and compute demand    |
| Process-aware  | Variant, state, alignment moves | Uses domain structure       | Model-dependent bias       |
| Graph          | DFG/object graph features       | Captures relationships      | Complex validation         |
| Object-centric | Features by object type         | Avoids premature flattening | Aggregation choices remain |

**Discuss** Start with an aggregation baseline. Add a richer encoding only when it improves validation utility enough to justify complexity.

# Feature provenance is part of the model

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For every feature, record:

- ▶ Source event or object attribute
- ▶ Aggregation and missing-value rule
- ▶ Earliest time at which the feature is available
- ▶ Case/object population to which it applies
- ▶ Fitted preprocessing parameters and training period

## Process-aware examples

Rework count, elapsed time, current activity, variant prefix, alignment cost, model state, handover count, open-object count, and workload at arrival.

# Responsible resource analysis

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- ▶ Analyze roles or teams before individuals where possible
- ▶ Check workload, case complexity, and data coverage
- ▶ Avoid ranking people using unvalidated process metrics
- ▶ Distinguish process constraints from individual performance
- ▶ Apply access controls, minimization, and purpose limitation

**Key idea** Event data reflects the system that assigns work. It is rarely a fair standalone measure of an individual's contribution.

# Correlation is a clue, not a cause

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A team may appear slower because it receives:

- ▶ More complex cases
- ▶ Cases already delayed upstream
- ▶ Work with poorer data quality
- ▶ Cases requiring additional controls

**Discuss** What additional data or study design would help distinguish a team effect from case-mix differences?